

Appendix

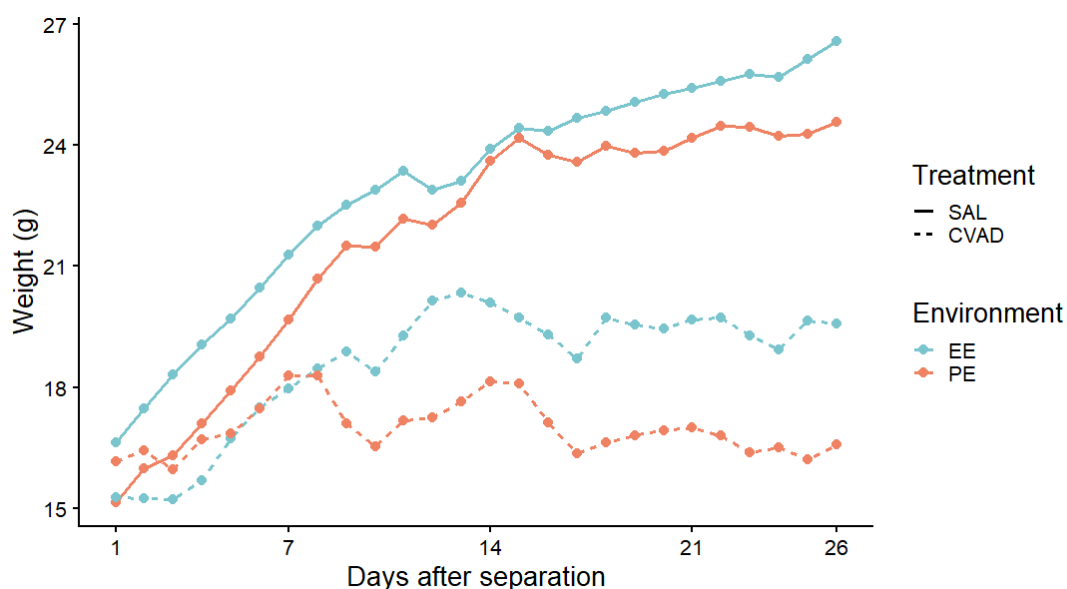
1. Supplementary analysis

1.1. Weights

Weight trajectories were visualized by plotting group mean body weight across days after separation, with cubic spline interpolation applied to weekend gaps. The pattern reveals a pronounced treatment effect: CVAD-treated mice show an early plateau in weight gain, in contrast to the continued growth observed in saline-treated controls.

Figure I

Weight curves per condition.



Note. Weight (g) plotted as a function of days after separation. Data beyond the treatment period are not shown, as weights were only assessed during treatment.

To formally evaluate the visual pattern, we fitted a mixed-effects model with random intercepts to the weight data (Table I). The analysis revealed a significant main effect of Day, indicating that all mice gained weight over time. There was no significant main effect of treatment or environment. The Day $\times$ Treatment interaction was significant, confirming that weight trajectories differed between CVAD- and saline-treated mice. Finally, the three-way interaction (Day $\times$ Treatment $\times$ Environment) indicated that environmental enrichment exerted a protective effect on weight gain over time, particularly for chemotherapy-treated mice. Bayesian analysis provided similar results (Table II).

Table I*Frequentist results from the weight analysis.*

	<i>t</i>	df	<i>p</i>	<i>R</i> ²
<i>Day</i>	32.27	871	<0.001***	0.21
<i>Environment</i>	-1.34	33	0.19	0.01
<i>Treatment</i>	-1.51	33	0.14	0.01
<i>Day x Environment</i>	0.74	871	0.46	<0.001
<i>Day x Treatment</i>	-10.80	871	<0.001***	0.03
<i>Environment x Treatment</i>	1.20	33	0.24	0.008
<i>Day x Treatment x environment</i>	-7.14	871	<0.001***	0.01

Note. $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***).**Table II***Bayesian results from the weight analysis.*

	<i>BF</i>	HDI low	HDI high
<i>Day</i>	>100	0.08	0.09
<i>Environment</i>	0.50	-0.70	0.10
<i>Treatment</i>	0.75	-0.84	0.07
<i>Day x Environment</i>	0.005	-0.003	0.009
<i>Day x Treatment</i>	>100	-0.05	-0.04
<i>Environment x Treatment</i>	0.70	-0.24	1.05
<i>Day x Treatment x Environment</i>	>100	-0.05	-0.03

Note. HDI low and HDI high indicate the bounds of the 90% highest-density interval.**1.2. SPSN**

To assess differences across the three SPSN stages, we fitted a repeated-measures ANOVA with stage as the within-subject factor and environment and treatment as fixed between-subject effects. The full results are shown in Table III.

Table III*Supplementary repeated-measures analysis of SPSN stages.*

	<i>F</i>	<i>p</i>	η^2
Between-subjects effects			
<i>treatment</i>	10.55	0.003**	0.26
<i>environment</i>	1.40	0.25	0.04
<i>treatment x env</i>	0.001	0.98	<0.001
Within-subjects effects			
<i>stage</i>	16.37	<0.001***	0.35
<i>treatment x stage</i>	1.68	0.20	0.05
<i>env x stage</i>	0.02	0.90	<0.001
<i>Treatment x stage x env</i>	1.48	0.23	0.05

Note. $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***).

Given the high p -value for Stage, the SPSN phases differed sufficiently that they could not be treated as originating from the same underlying mechanism. For this reason, we focused on the habituation phase when assessing general exploration, as it is the least influenced by social behavior or other processes.

Table IV*Frequentist results from the two-way ANOVA analysis of the social novelty preference.*

	<i>F</i>	<i>p</i>	η^2
<i>treatment</i>	0.20	0.66	0.007
<i>environment</i>	0.21	0.65	0.007
<i>treatment x env</i>	0.03	0.85	0.001

Note. $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***).

1.3. *Morris water maze*

Table V

Frequentist results from supplementary analysis of the acquisition phase (distance to target).

	<i>t</i>	df	<i>p</i>	<i>R</i> ²
<i>Day</i>	-4.30	1283	<0.001***	0.01
<i>Environment</i>	1.47	85	0.15	<0.001
<i>Treatment</i>	2.9	85	0.005**	0.003
<i>Day x Environment</i>	-3.35	1283	<0.001***	0.001
<i>Day x Treatment</i>	-1.51	1283	0.13	<0.001
<i>Environment x Treatment</i>	-2.58	85	0.01*	0.001
<i>Day x Treatment x environment</i>	1.82	1283	0.07	0.002

Note. $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***).

Table VI

Bayesian results from supplementary analysis of the acquisition phase (distance to target).

	<i>BF</i>	HDI low	HDI high
<i>Day</i>	20	-0.09	-0.04
<i>Environment</i>	0.44	-0.06	0.51
<i>Treatment</i>	9.65	0.26	1.11
<i>Day x Environment</i>	4.42	-0.10	-0.03
<i>Day x Treatment</i>	0.07	-0.09	0.01
<i>Environment x Treatment</i>	4.59	-1.27	-0.21
<i>Day x Treatment x Environment</i>	0.15	-0.001	0.13

Note. HDI low and HDI high indicate the bounds of the 90% highest-density interval.

Table VII

Frequentist results from supplementary analysis of the reversal acquisition phase.

	<i>t</i>	df	<i>p</i>	R^2
<i>Day</i>	-4.98	623	<0.001***	0.03
<i>Environment</i>	0.98	652	0.33	0.001
<i>Treatment</i>	0.62	652	0.54	0.001
<i>Day x Environment</i>	-1.20	623	0.23	0.002
<i>Day x Treatment</i>	-0.20	623	0.84	<0.001
<i>Environment x Treatment</i>	-1.07	652	0.29	0.002
<i>Day x Treatment x environment</i>	0.77	623	0.44	0.001

Note. $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***).

Table VIII

Bayesian results from supplementary analysis of the reversal acquisition phase.

	<i>BF</i>	HDI low	HDI high
<i>Day</i>	>100	-0.24	-0.12
<i>Environment</i>	0.71	-1.36	0.79
<i>Treatment</i>	1.43	-1.70	0.18
<i>Day x Environment</i>	0.06	-0.11	0.06
<i>Day x Treatment</i>	0.13	-0.01	0.13
<i>Environment x Treatment</i>	0.92	-1.67	0.71
<i>Day x Treatment x Environment</i>	0.06	-0.09	0.10

Note. HDI low and HDI high indicate the bounds of the 90% highest-density interval.

2. Bayesian models

This section contains all parameter-recovery plots and prior derivations for the Bayesian models used in the analysis. Parameter recovery was performed on simulated datasets generated from the specified priors.

2.1. Priors

When we conduct a two-way ANOVA in a Bayesian framework, we can express it as a linear model with dummy-coded predictors ($x_1, x_2 \in \{0,1\}$) and estimate the effects directly from the posterior:

$$y_{ij} = \mu + \alpha x_{1i} + \beta x_{2i} + \gamma(x_{1i}x_{2i}) + \epsilon_{ij} \quad \text{with } \epsilon_{ij} \sim N(0, \sigma)$$

Where i indexes subject and j indexes trial, α represents the weight of treatment, β the weight of environment and γ the interaction weight. This structure has the convenient property that each condition mean can be written as a combination of the model parameters. If we code environment as 0 = PE and 1 = EE, and treatment as 0 = Saline and 1 = CVAD, then the four

condition means become:

<i>Environment</i>	<i>Treatment</i>	
	<i>Saline</i>	<i>CVAD</i>
<i>PE</i>	μ	$\mu + \alpha$
<i>EE</i>	$\mu + \beta$	$\mu + \alpha + \beta + \gamma$

As becomes clear from the table, α and β represent simple effects. Specifically, α reflects how much better CVAD performs than Saline *within the PE condition*, and β reflects how much better EE performs than PE *within the Saline condition*. If we know that a main effect in an ANOVA is defined as the average effect of a factor across the levels of the other factor, then this implies for treatment that:

$$ME_t = \frac{\text{mean}(CVAD) - \text{mean}(SAL)}{2}$$

This formulation of an ANOVA-style main effect can again be rewritten as a combination of parameters:

$$ME_t = \frac{(2(\mu + \alpha) + \beta + \gamma) - (2\mu + \beta)}{2}$$

Or simplified:

$$ME_t = \alpha + \frac{\gamma}{2}$$

Using the same logic as before, we can apply this to the environment and interaction effects:

$$ME_{env} = \beta + \frac{\gamma}{2}$$

$$IE = \gamma$$

In this scenario, a Bayesian framework has a clear advantage. Because we obtain full posterior distributions for each parameter, we can apply any transformation directly to the posterior samples to derive the posterior of the transformed effect. This means that the posterior distributions of the ANOVA main effects can be obtained simply by transforming the posteriors of α , β , and γ accordingly. The difficulty arises when computing the Savage–Dickey density ratio, which requires not only the posterior density at zero but also the *prior* density of the same transformed effect. Once we transform the parameters to obtain a new posterior, the corresponding prior is no longer the original prior on the individual parameter but the induced prior on the transformed contrast. Suppose the priors for the model parameters are:

$$\alpha, \beta, \gamma \sim N(\theta, \sigma_p)$$

Because the transformation of the priors is linear, the induced prior then becomes:

$$ME_t, ME_{env} \sim N(\theta, \sigma_p) + 0.5 * N(\theta, \sigma_p)$$

$$ME_t, ME_{env} \sim N(\theta, \sigma_p) + N(0.5 * \theta, 0.25 * \sigma_p^2)$$

$$ME_t, ME_{env} \sim N(2\theta, \sigma_p + 0.25 * \sigma_p^2)$$

Given that the priors on our standardized effects are $N(0, I)$, we obtain the following prior distributions for the parameters and effects in our two-way ANOVA (where the same logic was applied to the contrasts):

Table IX

Priors for all two-way ANOVA.

	Priors
Regular effects	
μ	N(0,1)
α	N(0,1)
β	N(0,1)
γ	N(0,1)
σ_{res}	N(0,1)
ME_t	N(0,1.25)
ME_{env}	N(0,1.25)
Post-hoc comparisons	

<i>SAL diff</i>	N(0,1)
<i>CVAD diff</i>	N(0,2)
<i>PE diff</i>	N(0,1)
<i>EE diff</i>	N(0,2)
<i>Interaction diff</i>	N(0,3)

Note. μ is the grand mean, α is the simple treatment effect, β is the simple environment effect, and γ is the interaction effect. ME_t represents the main effect of treatment, while ME_{env} represents the main effect of environment. The post-hoc comparison priors correspond to the induced priors for each pairwise contrast: environment differences conditioned on a treatment (SAL diff, CVAD diff), treatment differences conditioned on each environment (PE diff, EE diff), and the SAL PE to CVAD EE difference (Interaction diff).

For all linear mixed-effects models, the same priors for all parameters were used:

Table X

Priors for all the mixed-effects linear models.

	Priors
<i>Simple effects ($\beta_t, \beta_{env}, \beta_{day}$)</i>	N(0,1)
<i>Two-way interactions ($\beta_{e_t}, \beta_{d_e}, \beta_{d_t}$)</i>	N(0,1)
<i>Three-way interaction</i>	N(0,1)
<i>Population mean (μ)</i>	N(0,1)
<i>Population variance (σ_{group})</i>	N(0,1)
<i>Residual variance (σ_{res})</i>	N(0,1)

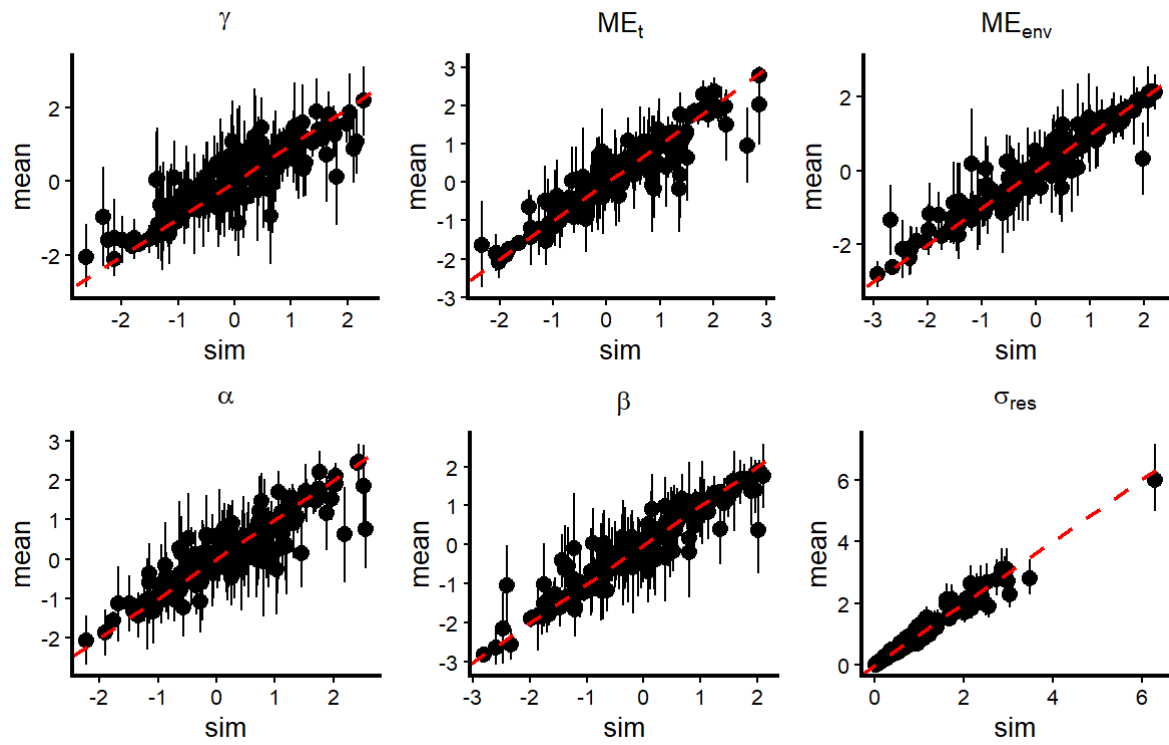
Note. β_{day} = day weight; β_{env} = environment weight; β_t = treatment weight; β_{d_e} = day x environment weight; β_{d_t} = day x treatment weight; β_{e_t} = environment x treatment weight; β_{int} = three-way interaction; μ = group mean intercept; σ_{res} = residual variance.

2.2. Parameter recovery

All parameter recoveries were run on simulated datasets and fitted using the same models applied in the main analysis. We generated 200 datasets by sampling a true value for each model parameter and then attempted to recover these values by fitting the simulated data back to the model. The following plots show the simulated (true) parameter values alongside the estimated values and their 90% HDIs.

Figure II

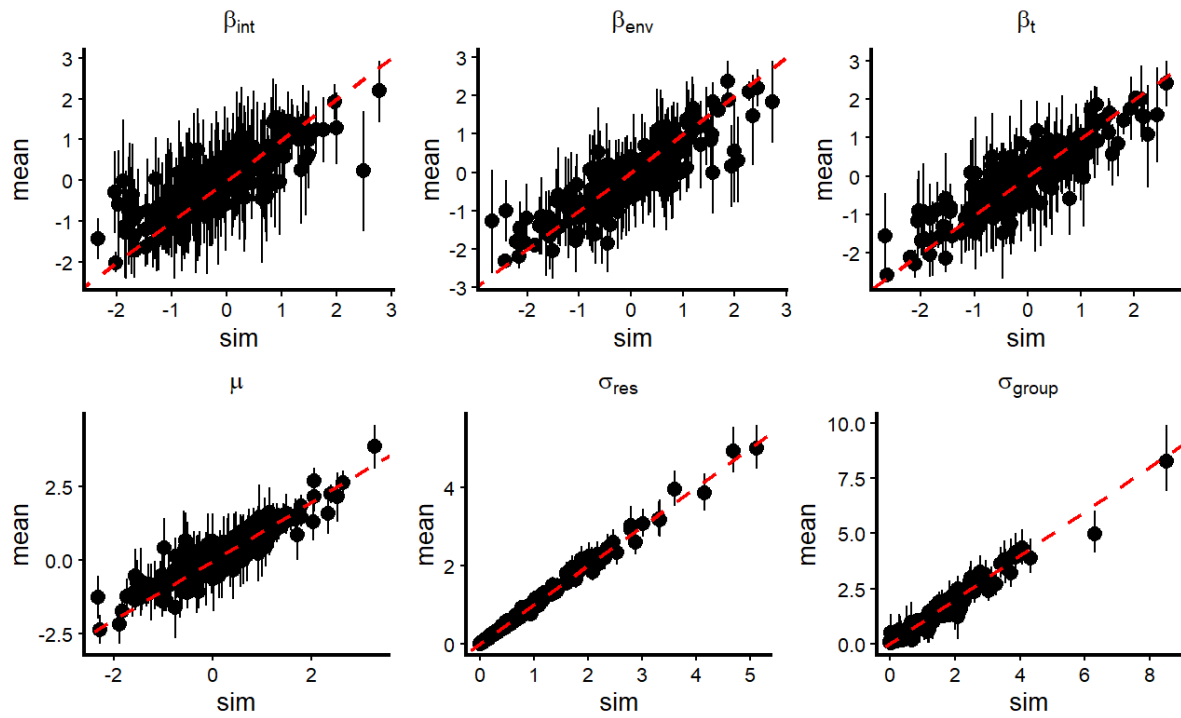
Parameter recovery for the two-way ANOVA.



Note. γ is the interaction effect, ME_t is the main effect of treatment, and ME_{env} the main effect of environment. α is the simple treatment effect and β is the simple environment effect. σ_{res} is the residual variance.

Figure III

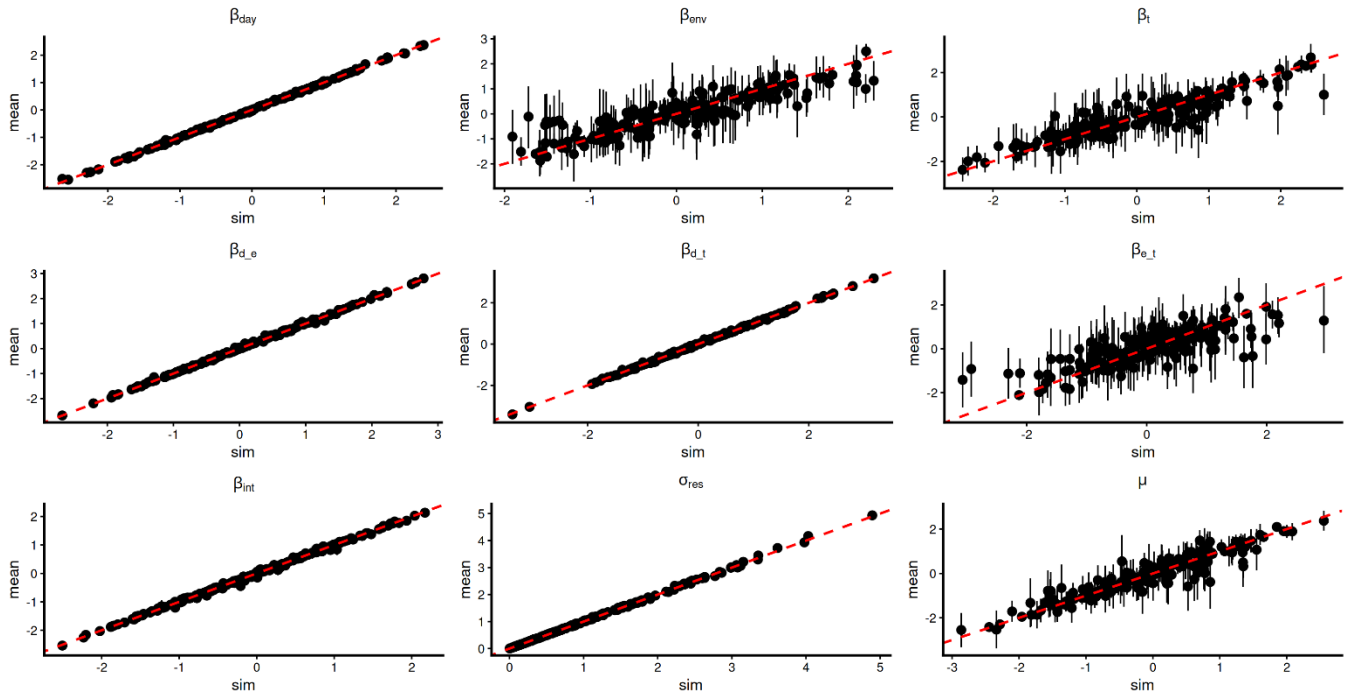
Parameter recovery for the 2-factor mixed-effects model with random intercepts.



Note. β_{int} = interaction weight; β_{env} = environment weight; β_t = treatment weight; μ = group mean intercept; σ_{res} = residual variance; σ_{group} = the variance around group mean. Note that the environment and treatment weights are not equivalent to ANOVA main effects, because the predictors were dummy-coded.

Figure IV

Parameter recovery for the 3-factor mixed-effects model with random intercepts.



Note. β_{day} = day weight; β_{env} = environment weight; β_t = treatment weight; β_{d_e} = day x environment weight; β_{d_t} = day x treatment weight; β_{e_t} = environment x treatment weight; β_{int} = three-way interaction; μ = group mean intercept; σ_{res} = residual variance.

3. Additional material

Figure V

Humane endpoints applied in this study.

Sign		Action	
Body weight			
Adults	<10-15% within 24h	Check mouse: teeth ok? Place food in cage, check if water bottle is clogged provide peanut butter in cage give saline i.p. (up to 5% bw)	Monitor until bw is back to normal
	<20%	Check mouse: teeth ok? Place food in cage, check if water bottle is clogged provide peanut butter in cage give saline i.p. (up to 5% bw)	Check 24h later again, if not improved, kill animal
Pups, juvenile	<10%	Place food in cage, check if water bottle is clogged provide peanut butter in cage give saline i.p. (up to 5% bw)	Check 24h later again, if not improved, kill animal
General signs			
Adults	Piloerection	>5 days	Kill
	Hunched posture	>5 days	Kill
	Subdued behavior	Subdued even if provoked >2d	Kill
	Stereotypic behavior	Cannot be interrupted and affects health and welfare	Kill
	Trembling	>5d	Kill
	Dehydration	Check if water bottle is ok	Provide s.c. hydration (saline) 5% bw, >5d kill
	Vocalization	Distressed or unprovoked vocalization	Kill
Mobility	Single limb	>5d, if welfare is affected/pain is visible	Kill
	Abnormal mobility (slow, stumbling, circling)	>5d	Kill
Wounds	Bite wounds, etc	>10 mm (body), >3mm face	Kill
		<10mm	Put some antibiotic cream/vaseline on it to heal
Social behavior	Abnormal isolation	Avoiding group	Isolate in a new cage

Note. This figure presents the exact humane endpoints displayed throughout the laboratory. Endpoints are organised by behavioural signs, the checks required, and the corresponding actions to be taken.